3. Influx of phagocytes from the capillaries into the tissues is facilitated by increased capillary permeability. As phagocytic cells accumulate at the site and begin to phagocytoses bacteria, they release lytic enzymes, which can damage nearby healthy cells. The accumulation of dead cells, digested material and fluid forms substances called pus.

13.4.3 Acquired Immunity

Lower animal forms possess so called innate or non-specific immune mechanisms such as phagocytosis of bacteria by specialized cells. Higher animals have evolved an adaptive or acquired immune response. This acquired immune response provides a flexible, specific and more effective reaction to different infections.

- **Definition of Acquired (Adaptive) Immunity**

Acquired (adaptive)immunity refers to the type of specific immunity that a host develops after exposure to a suitable antigen.

- **Important features of acquired immunity**

This is the immunity one develops throughout life time. Adaptive or acquired immunity has four important features namely (1) Memory (2) Specificity (3) diversity and (4) discrimination between self and non self.

1) Memory

We rarely suffer twice from diseases such as measles, mumps, chicken pox, whooping cough and so on. The first contact with an infectious organism clearly imprints some memory so that the body is effectively prepared to repel any later invasion by that organism.

2) Specificity

The establishment of immunity by one organism does not provide protection against another unrelated organism. After an attack of measles we are immune to further infection but are susceptible to polio or mumps viruses. Thus the body can differentiate specifically between the two organisms.

3) Diversity

The immune system is able to generate an enormous diversity of molecules such as cellular receptors and soluble proteins, including antibodies that recognize trillions of different foreign substances.

4) Discrimination between self and nonself

The specific immune system almost responds selectively to non self and produces specific responses against the stimulus. This is possible because host cells express a unique protein on their surface, making them as residents of that host or as self. Thus the introduction of materials lacking that unique self marker results in their attack by the host.

13.4.4 Humoral and Cellular Immunity

Two branches or arms of specific immunity are recognized: humoral (antibody mediated) immunity and cellular (cell mediated) immunity (Figure 13.20).

Humoral (antibody mediated) immunity

The antigen specific arm of the humoral immunity consists of the B cells. Each B cell expresses a unique antigen binding receptor on its membrane. The B cell receptor (BCR) is membrane bound antibody molecule. When a naive B cell first encounters the antigen that matches its membrane bound

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